Random Processes

Chapter 2: Practice set 2 — coupling and limiting distributions

University of Michigan

Scope. Part A: total variation and couplings. Part B: existence and computation of limiting distributions. Result numbers refer to the Chapter 2 notes (Theorem 6.7 = limit theorems, Lemma 7.5 = coupling inequality, Theorem 7.22 = maximal coupling). A $\star$ on a problem or a part means it is harder than the Fall 2026 practice midterm; unmarked problems are at about exam level. Solutions start on a new page after the problems.

Part A: Coupling and total variation

Problem 1 (TV and a maximal coupling).
Let $\mu = \mathrm{Bin}(2, \frac{1}{2})$ and $\nu = \mathrm{Unif}\{0, 1, 2\}$.

(a) Compute $\|\mu - \nu\|_{\mathrm{TV}}$.

(b) Write the 3×3 joint table of a coupling (U, V) with $U \sim \mu$, $V \sim \nu$ and $\mathbb{P}(U \neq V) = \|\mu - \nu\|_{\mathrm{TV}}$.

Problem 2 (Two binomials).
Let $0 \leq p \leq q \leq 1$. Using a coupling of n pairs of Bernoullis, show

$$\|\mathrm{Bin}(n, p) - \mathrm{Bin}(n, q)\|_{\mathrm{TV}} \leq n(q - p).$$

Problem 3 ⋆ (Doeblin coupling).
Let P be a transition matrix on a finite S, and put

$$\delta = \sum_{y \in S} \min_{x \in S} p(x, y).$$

Assume $\delta > 0$.

(a) Define $\nu(y) = \min_x p(x, y)/\delta$ and, if $\delta < 1$, $R(x, y) = (p(x, y) - \delta\nu(y))/(1 - \delta)$. Show that ν is a probability vector and R is a transition matrix.

(b) Build a coupling of two P-chains as follows. At each step, with probability δ both chains jump to a common state drawn from ν; otherwise, each moves by R (independently if they differ, together if they are equal). Check that each coordinate is a P-chain, and deduce

$$\|P^n(x, \cdot) - P^n(y, \cdot)\|_{\mathrm{TV}} \le (1 - \delta)^n, \quad \|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \le (1 - \delta)^n.$$

(c) Apply this to

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \\ 0.3 & 0.2 & 0.5 \end{pmatrix}.$$

What is π? What bound do you get for $n = 5$?

Problem 4 ⋆ (Lazy walk on a cycle).
Let $m \ge 3$ and let P be the lazy walk on $\mathbb{Z}_m$: $p(x, x) = \frac{1}{2}$, $p(x, x \pm 1) = \frac{1}{4}$.

(a) Start $X_0 = x$, $Y_0 = y$. Until they meet, flip a fair coin each step: on heads X moves to $X \pm 1$ (each with probability $\frac{1}{2}$) and Y stays; on tails Y moves and X stays. After meeting, they move together. Show each coordinate is a lazy walk.

(b) Show that $D_n = X_n - Y_n \pmod m$ performs simple random walk until it hits 0, and use gambler's ruin to show $\mathbb{E}\tau \le m^2/4$.

(c) Deduce $\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \le m^2/(4n)$. How large must n be to guarantee distance at most $\frac{1}{4}$?

Problem 5 ⋆ (Lazy walk on the hypercube).
On $S = \{0, 1\}^d$, one step picks a coordinate i uniformly and a fair bit B, and sets $x_i = B$.

(a) Show the chain is irreducible and aperiodic with uniform stationary distribution.

(b) Couple two copies using the same (i, B) at every step. Show $\tau \le T$, where T is the coupon-collector time for the d coordinates, and deduce $\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \le de^{-n/d}$.

(c) For $d = 10$, how many steps guarantee distance at most 0.01?

Problem 6 ⋆ (Monotone coupling).
Reflecting walk on $\{0,1,\dots,N\}$: $p(x,x+1)=p$ for $x<N$, $p(x,x-1)=q=1-p$ for $x>0$, $p(0,0)=q$, $p(N,N)=p$.

(a) Write one step as $X_{j+1}=f(X_j,U_{j+1})$ with $U_{j+1}\sim\mathrm{Unif}[0,1]$ and $f(x,u)$ nondecreasing in x.

(b) Drive two chains from $x\le y$ with the same U's. Show $X_n\le Y_n$ for all n, and deduce that $x\mapsto\mathbb{P}_x(X_n\ge k)$ is nondecreasing for every n,k.

(c) Show the gap Y_n-X_n never increases, and that the chains are equal from the first time X hits N or Y hits 0.

Part B: Limiting distributions

Problem 7 (Which chains have a limiting distribution?).
For each matrix on $S=\{1,2,3\}$, decide whether a limiting distribution exists. If it does, find it; if not, say what goes wrong and compute whatever limit does exist (row limits or Cesàro limits).

$$P_1=\begin{pmatrix}0&\frac12&\frac12\\1&0&0\\1&0&0\end{pmatrix},\quad P_2=\begin{pmatrix}\frac12&\frac12&0\\\frac14&\frac34&0\\\frac13&\frac13&\frac13\end{pmatrix},\quad P_3=\begin{pmatrix}1&0&0\\0&1&0\\\frac12&\frac14&\frac14\end{pmatrix},\quad P_4=\begin{pmatrix}0&1&0\\1&0&0\\\frac12&0&\frac12\end{pmatrix}.$$

Problem 8 (Absorption into closed classes).
On $S=\{1,\dots,5\}$ let

$$P=\begin{pmatrix}\frac12&\frac12&0&0&0\\\frac13&\frac23&0&0&0\\0&0&1&0&0\\\frac14&0&\frac14&\frac14&\frac14\\0&\frac12&\frac14&\frac14&0\end{pmatrix}.$$

(a) Classify the states and find the closed classes.

(b) Find $\alpha(x)=\mathbb{P}_x(\text{enter }\{1,2\})$ for $x=4,5$.

(c) Compute $\lim_n P^n(4,\cdot)$ and $\lim_n P^n(5,\cdot)$. Is there a limiting distribution?

Problem 9 (Success-run chain).
On $S=\mathbb{Z}_{\ge0}$, fix $p\in(0,1)$, $q=1-p$, and let $p(i,i+1)=p$, $p(i,0)=q$ for every i.

(a) Show the chain is irreducible and aperiodic.

(b) Find the stationary distribution, $\lim_n P^n(x,y)$, and $\mathbb{E}_0T_0$.

(c) What is the long-run fraction of time the chain spends at states ≥3?

(d) ⋆ Couple two copies from x and y by using the same coin at every step (both go up, or both reset to 0). Show $\|P^n(x,\cdot)-\pi\|_{\mathrm{TV}}\le p^n$.

Problem 10 (Ehrenfest urn).
m balls are split between two urns; at each step a ball is chosen uniformly and moved to the other urn. Let X_n be the number of balls in urn 1, so $p(i, i-1) = \frac{i}{m}$ and $p(i, i+1) = \frac{m-i}{m}$.

(a) Find the stationary distribution using detailed balance.

(b) Find the period. Does $P^n(0, \cdot)$ converge? Compute $\lim_n P^{2n}(0, y)$ for $m = 4$.

(c) For the lazy chain $Q = (I + P)/2$, find $\lim_n Q^n(x, y)$.

(d) Starting from an empty urn 1, what is the expected time to return to an empty urn 1? Evaluate for $m = 10$.

(e) $\star$ Show that the number of ones in the hypercube walk of Problem 5 (with $d = m$) is exactly the lazy Ehrenfest chain Q. Deduce $\|Q^n(0, \cdot) - \pi\|_{\mathrm{TV}} \le m e^{-n/m}$.

Problem 11 (Walks on cycles: aperiodic or not?).

(a) On $\mathbb{Z}_5$ let $p(x, x+1) = p(x, x+2) = \frac{1}{2}$. Show the chain is irreducible and aperiodic, and find $\lim_n P^n(0, y)$.

(b) On $\mathbb{Z}_6$ let $p(x, x+1) = p(x, x+3) = \frac{1}{2}$. Show the chain is irreducible with period 2. Is there a limiting distribution?

(c) $\star$ For the chain in (b), compute $\lim_n P^{2n}(0, y)$ and $\lim_n P^{2n+1}(0, y)$.

Solutions

Solution 1.
(a) $\mu = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$ and $\nu = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. The overlap is $r = \min(\mu, \nu) = \left(\frac{1}{4}, \frac{1}{3}, \frac{1}{4}\right)$ with $\sum r = \frac{5}{6}$, so $\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{6}$.

(b) The excess of μ is $\frac{1}{6}$ at 1; the excess of ν is $\frac{1}{12}$ at 0 and $\frac{1}{12}$ at 2. Following Theorem 7.22, put r on the diagonal and send μ's excess at 1 to ν's excess at 0 and 2 (rows: U; columns: V):

	$V=0$	$V=1$	$V=2$
$U=0$	$\frac{1}{4}$	0	0
$U=1$	$\frac{1}{12}$	$\frac{1}{3}$	$\frac{1}{12}$
$U=2$	0	0	$\frac{1}{4}$

Row sums $\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right) = \mu$, column sums $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \nu$, and $\mathbb{P}(U \neq V) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$.

Solution 2.
Take $Z_1, ..., Z_n$ iid uniform on $[0, 1]$ and put $U_i = \mathbf{1}\{Z_i \leq p\}$, $V_i = \mathbf{1}\{Z_i \leq q\}$. Then $\sum U_i \sim \mathrm{Bin}(n, p)$, $\sum V_i \sim \mathrm{Bin}(n, q)$, and $\mathbb{P}(U_i \neq V_i) = \mathbb{P}(p < Z_i \leq q) = q - p$. Since $\sum U_i \neq \sum V_i$ forces some $U_i \neq V_i$, the coupling inequality and a union bound give

$$\|\mathrm{Bin}(n, p) - \mathrm{Bin}(n, q)\|_{\mathrm{TV}} \leq \mathbb{P}\left(\sum_i U_i \neq \sum_i V_i\right) \leq \sum_i \mathbb{P}(U_i \neq V_i) = n(q - p).$$

Solution 3.
(a) $\nu \geq 0$ and $\sum_y \nu(y) = \delta/\delta = 1$. For every x, y, $p(x, y) \geq \min_{x'} p(x', y) = \delta\nu(y)$, so $R \geq 0$, and $\sum_y R(x, y) = (1 - \delta)/(1 - \delta) = 1$.

(b) From x, the next state is y with probability $\delta\nu(y) + (1 - \delta)R(x, y) = p(x, y)$, whether or not the chains have met; so each coordinate is a P-chain. Each step, independently of the past, the "common jump" happens with probability δ, and after it the chains are equal forever. Hence $\mathbb{P}(\tau > n) \leq (1 - \delta)^n$, and the coupling inequality gives the first bound. For the second, by convexity,

$$\begin{aligned}\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} &= \left\|\sum_y \pi(y)(P^n(x, \cdot) - P^n(y, \cdot))\right\|_{\mathrm{TV}} \\ &\leq \sum_y \pi(y)\|P^n(x, \cdot) - P^n(y, \cdot)\|_{\mathrm{TV}} \leq (1 - \delta)^n.\end{aligned}$$

(c) Every column of P sums to 1, so P is doubly stochastic and $\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. Column minima are $0.2, 0.2, 0.2$, so $\delta = 0.6$ and the bound is 0.4^n; for $n = 5$ it is $0.4^5 \approx 0.0102$. (The true distance from state 1 at $n = 5$ is about 0.0008, so the bound is valid but not tight.)

Solution 4.
(a) On heads (probability $\frac{1}{2}$) X moves to $X \pm 1$ with probability $\frac{1}{2}$ each; on tails it stays. So X stays with probability $\frac{1}{2}$ and moves to each neighbour with probability $\frac{1}{4}$: a lazy walk. The same holds for Y, and after meeting they take a common lazy step.

(b) On heads D changes by ± 1 (from X's move), on tails by ∓ 1 (from Y's move); either way $D_{n+1} = D_n \pm 1$ with probability $\frac{1}{2}$ each. So until meeting, D is simple random walk on $\mathbb{Z}_m$, and τ is the time it hits 0, i.e. gambler's ruin on $\{0, ..., m\}$ absorbed at 0 or m. From $D_0 = k$, $\mathbb{E}\tau = k(m-k) \le m^2/4$ (fair gambler's ruin, Lecture 8).

(c) By the coupling inequality and Markov's inequality,

$$\|P^n(x,\cdot) - P^n(y,\cdot)\|_{\mathrm{TV}} \le \mathbb{P}(\tau > n) \le \frac{\mathbb{E}\tau}{n} \le \frac{m^2}{4n}.$$

Average over $y \sim \pi$ as in Solution 3(b) to get $\|P^n(x,\cdot) - \pi\|_{\mathrm{TV}} \le m^2/(4n)$. Distance at most $\frac{1}{4}$ is guaranteed once $n \ge m^2$.

Solution 5.

(a) Changing coordinates one at a time reaches any y from any x, so the chain is irreducible. $p(x,x) = \frac{1}{2}$ (whatever i is, $B = x_i$ with probability $\frac{1}{2}$), so it is aperiodic. For $y \ne x$, $p(x,y) = 1/(2d)$ if x, y differ in exactly one coordinate and 0 otherwise; P is symmetric, hence doubly stochastic, and π is uniform: $\pi(x) = 2^{-d}$.

(b) With shared (i, B), coordinate i is set to the same bit in both chains and agrees forever after. Once every coordinate has been chosen at least once (time T), the chains are equal, so $\tau \le T$. Then

$$\|P^n(x,\cdot) - \pi\|_{\mathrm{TV}} \le \mathbb{P}(T > n) \le \sum_{i=1}^{d} \mathbb{P}(i \text{ not chosen by time } n) = d\left(1 - \frac{1}{d}\right)^n \le de^{-n/d}.$$

(c) $de^{-n/d} \le 0.01$ iff $n \ge d(\log d + \log 100) = 10(2.303 + 4.605) \approx 69.1$, so $n = 70$ steps suffice.

Solution 6.

(a) $f(x,u) = \min(x+1, N)$ if $u \le p$ and $f(x,u) = \max(x-1, 0)$ if $u > p$. For fixed u, both branches are nondecreasing in x, and $f(x, U)$ has the right transition probabilities.

(b) If $X_j \le Y_j$ then $X_{j+1} = f(X_j, U_{j+1}) \le f(Y_j, U_{j+1}) = Y_{j+1}$; induct from $X_0 = x \le y = Y_0$. Hence $\{X_n \ge k\} \subseteq \{Y_n \ge k\}$, so

$$\mathbb{P}_x(X_n \ge k) = \mathbb{P}(X_n \ge k) \le \mathbb{P}(Y_n \ge k) = \mathbb{P}_y(Y_n \ge k).$$

(c) Both chains try the same move. Away from the boundary both move by the same ± 1, so the gap is unchanged. The gap can only change when one chain is blocked: Y at N trying to go up (gap shrinks by 1) or X at 0 trying to go down (gap shrinks by 1); it never grows. If X hits N, then $N \ge Y_n \ge X_n = N$, so they are equal; similarly if Y hits 0. Once equal, the same U's keep them equal.

Solution 7.

P_1: irreducible ($1 \leftrightarrow 2$, $1 \leftrightarrow 3$) with period 2 (bipartite: $\{1\}$ versus $\{2,3\}$). Stationary $\pi = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$, but $P_1^{2n}(1,\cdot) = (1,0,0)$ and $P_1^{2n+1}(1,\cdot) = \left(0, \frac{1}{2}, \frac{1}{2}\right)$: no limiting distribution. The Cesàro limit is π.

P_2: $\{1,2\}$ is closed and aperiodic, 3 is transient. Exactly one closed class, aperiodic, so a limiting distribution exists. On $\{1,2\}$, $\pi(1) \cdot \frac{1}{2} = \pi(2) \cdot \frac{1}{4}$, so $\pi = \left(\frac{1}{3}, \frac{2}{3}, 0\right)$, and every row of P_2^n tends to it.

P_3: two absorbing states $1, 2$; 3 is transient. Rows 1 and 2 converge to δ_1 and δ_2, which differ, so there is no limiting distribution. From 3, $h = \mathbb{P}_3(\text{absorbed at 1})$ solves $h = \frac{1}{2} + \frac{h}{4}$, so $h = \frac{2}{3}$ and $P_3^n(3, \cdot) \to \left(\frac{2}{3}, \frac{1}{3}, 0\right)$.

P_4: the only closed class $\{1, 2\}$ has period 2, so $P_4^n(1, 1)$ alternates $0, 1$: no limiting distribution. Cesàro limits exist: every row averages to $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$.

Solution 8.

(a) $\{1, 2\}$ is closed and aperiodic (self-loops); 3 is absorbing; 4 and 5 are transient (they lead to 3, which never returns).

(b) $\alpha = \mathbb{P}(\text{enter } \{1, 2\})$ satisfies $\alpha_4 = \frac{1}{4} + \frac{1}{4}\alpha_4 + \frac{1}{4}\alpha_5$ and $\alpha_5 = \frac{1}{2} + \frac{1}{4}\alpha_4$. The first gives $3\alpha_4 = 1 + \alpha_5$; substituting, $\frac{11}{4}\alpha_4 = \frac{3}{2}$, so

$$\alpha_4 = \frac{6}{11}, \quad \alpha_5 = \frac{7}{11}.$$

(c) On $\{1, 2\}$, $\pi(1) \cdot \frac{1}{2} = \pi(2) \cdot \frac{1}{3}$, so $\pi_{\{1,2\}} = \left(\frac{2}{5}, \frac{3}{5}\right)$. By the mixture formula $\lim_n P^n(x, \cdot) = \alpha(x)\pi_{\{1,2\}} + (1 - \alpha(x))\delta_3$:

$$\lim_n P^n(4, \cdot) = \left(\frac{12}{55}, \frac{18}{55}, \frac{25}{55}, 0, 0\right), \quad \lim_n P^n(5, \cdot) = \left(\frac{14}{55}, \frac{21}{55}, \frac{20}{55}, 0, 0\right).$$

The rows differ (two closed classes), so there is no limiting distribution.

Solution 9.

(a) Every state reaches 0 in one step and 0 reaches i in i steps, so the chain is irreducible; $p(0, 0) = q > 0$, so it is aperiodic.

(b) $\pi(i + 1) = p\pi(i)$ and $\pi(0) = q\sum_i \pi(i) = q$, so $\pi(i) = qp^i$, a geometric distribution. I + A + S gives $\lim_n P^n(x, y) = qp^y$ for every x, and $\mathbb{E}_0 T_0 = 1/\pi(0) = 1/q$. (Directly: the return time to 0 is the first reset, which is geometric with mean $1/q$.)

(c) By Theorem 6.7(2)–(4), the fraction is $\sum_{i \geq 3} qp^i = p^3$.

(d) Under the shared coin, the first reset sends both chains to 0, after which they are equal. The first reset happens after time n with probability p^n, so $\|P^n(x, \cdot) - P^n(y, \cdot)\|_{\text{TV}} \leq p^n$; averaging over $y \sim \pi$ gives $\|P^n(x, \cdot) - \pi\|_{\text{TV}} \leq p^n$. This works on a countable state space with a bound uniform in x.

Solution 10.

(a) Detailed balance: $\pi(i)\frac{m-i}{m} = \pi(i + 1)\frac{i+1}{m}$, so $\frac{\pi(i+1)}{\pi(i)} = \frac{m-i}{i+1}$ and $\pi(i) = \binom{m}{i}2^{-m}$, the $\text{Bin}\left(m, \frac{1}{2}\right)$ distribution.

(b) Every step changes X_n by ± 1, and $0 \to 1 \to 0$ has length 2: period 2. $P^n(0, 0) = 0$ for odd n, so $P^n(0, \cdot)$ does not converge. From 0, even times land on even states; P^2 on the even states is irreducible and aperiodic with stationary law proportional to π, so $\lim_n P^{2n}(0, y) = 2\pi(y)$ for even y and 0 for odd y. For $m = 4$: $\left(\frac{2}{16}, 0, \frac{12}{16}, 0, \frac{2}{16}\right) = \left(\frac{1}{8}, 0, \frac{3}{4}, 0, \frac{1}{8}\right)$.

(c) Q has the same stationary π and $Q(x, x) = \frac{1}{2}$, so it is irreducible and aperiodic: $\lim_n Q^n(x, y) = \binom{m}{y}2^{-m}$ for every x.

(d) $\mathbb{E}_0 T_0 = 1/\pi(0) = 2^m$ (for P and for Q). For $m = 10$ this is 1024 steps, even though the walk starts next to the most likely states; this is why the urn "never" returns to empty in practice.

(e) In the hypercube walk, the chosen bit changes with probability $\frac{1}{2}$ (when $B \neq x_i$). Given k ones, it decreases with probability $\frac{1}{2} \cdot \frac{k}{m}$, increases with probability $\frac{1}{2} \cdot \frac{m-k}{m}$, and stays put with probability $\frac{1}{2}$: exactly Q. The number of ones is a function of the hypercube state, and TV cannot increase under a function, so $\|Q^n(0,\cdot) - \pi\|_{\text{TV}} \leq \|P^n_{\text{cube}}(\mathbf{0},\cdot) - \text{Unif}\|_{\text{TV}} \leq me^{-n/m}$ by Solution 5(b).

Solution 11.

(a) $x \to x+1$ connects all of $\mathbb{Z}_5$, so it is irreducible. Return lengths include 3 ($1+2+2=5$) and 4 ($1+1+1+2=5$), and $\gcd(3,4)=1$: aperiodic. Each column has two entries $\frac{1}{2}$, so P is doubly stochastic and π is uniform; Theorem 6.7(1) gives $\lim_n P^n(0,y) = \frac{1}{5}$.

(b) Again $x \to x+1$ gives irreducibility. Both steps are odd and 6 is even, so the parity of X_n flips at every step: returns to 0 need an even number of steps. Since $3+3=6$, a return in 2 steps is possible, so the period is 2. A periodic irreducible chain has no limiting distribution.

(c) π is uniform (P is doubly stochastic). As for Problem 2 of the first set, P^2 restricted to the even states $\{0,2,4\}$ is irreducible and aperiodic with uniform stationary law, so

$$\lim_n P^{2n}(0,\cdot) = \left(\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0\right), \quad \lim_n P^{2n+1}(0,\cdot) = \left(0, \frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}\right).$$